The Unequal Energy Divide - Power, Prosperity, and the Path to Clean Growth

Abstract

This project investigates how national economic development, electricity access, population size and renewable energy adoption interact to create global energy inequality. Using integrated datasets from Ember Energy (Yearly Electricity Data), the World Bank (Total Population) and Our World in Data (Urbanisation and GDP per Capita) spanning 2000–2023, the research develops a unified analytical model. A multidimensional bubble chart visualisation created in Tableau simultaneously represents economic prosperity, electricity consumption per capita, population size and clean energy adoption. A key feature of this visualisation is the 5,000 kWh per capita Energy Poverty Threshold, which distinguishes adequate from inadequate access.

The results reveal a persistent global gap. High-income, highly urbanised countries consistently exceed the threshold, while many low and middle-income nations lag significantly behind. However, a paradox emerges: several developing economies are making strides in renewable energy despite limited financial resources. This suggests that while economic prosperity strongly predicts electricity availability, it does not exclusively determine the pace of the clean energy transition.

The project emphasises the need for policies to address energy poverty, promote equitable renewable energy expansion and support fair global burden-sharing.

1. Datasets

This project relies on four key datasets, combined into one, Final_CountryOnly.csv, containing 313,918 rows and 23 columns, taking up 40.5 MB of space; this serves as the base for most of the analysis. These four datasets provide a comprehensive overview of global electricity generation, economic capacity, population, and urbanization.

Electricity Generation Dataset (Main)

Source: Ember Energy

Content: Annual electricity generation by country, including total output, fuel-type breakdown (fossil fuels, hydro, wind, solar, nuclear, other renewables), and CO_2 emissions.

Size: 357,076 rows × 18 columns (41.5 MB)

Key Attributes: Country, ISO code, Year, Area type, Fuel Type, Value, Unit, Continent, G20, EU.

- Data Types: String (country), integer (year), float (generation/output, CO_2).
- Big Data Aspects: Volume – high; Variety – high (categorical, time series, type differentiation); Velocity – low (annual release).

Population Dataset

- Source: Population
- Content: Annual total population for each country, used for per-capita calculations and bubble
- sizes.
- Size: 16,930 rows × 4 columns (525 KB)
- Key Attributes: Country, Country Code, Year, Population.
- Data Types: String (country, code), integer (year), float (population).
  Big Data Aspects: Volume – low; Variety – moderate; Velocity – low (annual batches).

Urban Population Dataset

- Source: Share of the population living in urban areas, 2024
- Content: Percentage of urban population per country; used to assess infrastructure readiness.
- Size: 14,820 rows × 4 columns (385 KB)
- Key Attributes: Country, Country Code, Year, Urban population (%).
- Data Types: String, integer, float.
- Big Data Aspects: Volume – low; Variety – moderate; Velocity – low.
- Note: Excluded from final visualization due to high correlation with GDP per capita.

GDP Dataset

- Source: GDP per capita - Our World in Data
- Content: GDP per capita (constant 2015 USD) to represent economic development in analysis.
- Size: 12,098 rows × 4 columns (345 KB)
- Key Attributes: Country, Country Code, Year, GDP per capita.
  Data Types: String, integer, float.
- Big Data Aspects: Volume – low; Variety – moderate; Velocity – low.

Together, these datasets allow for multivariate analysis of global electricity access, economic capacity, and population distribution, which serves as the basis for the bubble chart visualization.

2. Data Exploration, Processing, Cleaning, and Integration

2.1 Data Integration and Initial Exploration

We combined in our analysis four primary data sets Electricity Generation, Population, Urban Population, and GDP spanning 2000–2023 to study economic development and energy access. Each of the mentioned datasets were merged into one comprehensive dataset for cross-country and cross-variable analysis. Critical consistency checks were performed:

- **Temporal Alignment:** Ensured consistent year coverage.
- **Geographical Normalization:** Standardization of country names and codes.
- **Unit Consistency:** GDP in constant USD; electricity generation in TWh/GWh; renewable share as a percentage.

Urban Population (%) was included originally to see if any correlations existed with electricity access; however, in exploratory analysis, this was highly overlapped by GDP per capita, and thus excluded in the final visualisation to avoid redundancy and keep this clear.

2.2 Data Processing

Because data sizes were large, local processing in Jupyter Notebook using pandas was used. Key steps in this regard included:

- **Field Extraction:** Only relevant fields were kept, namely Year, Country, GDP, Population, Electricity Output, and Fuel Type.
- **Per-Capita Metrics:** calculated electricity generation per capita and renewable share for comparative analysis.
- **Chunk Processing:** In order to handle memory efficiently, large CSV files were processed in chunks.

2.3 Data Exploration, Validation, and Structuring

- **Handling Missing Values:** Records that had null GDP or electricity values were excluded to preserve accuracy.
- **Outlier Management:** Records that were extreme or improbable were modified or removed to prevent distortion of per-capita metrics.
- **Filtering and Grouping:** Countries were grouped into analytical groups such as G20, EU, ASEAN, and OECD, and also into GDP Income Groups like High, Middle, Low Income to avoid overplotting and provide interactive filtering in the bubble chart

The cleaned and integrated dataset was afterwards exported, optimized for visualization in a .csv file, Final_CountryOnly.csv, ready for analysis.

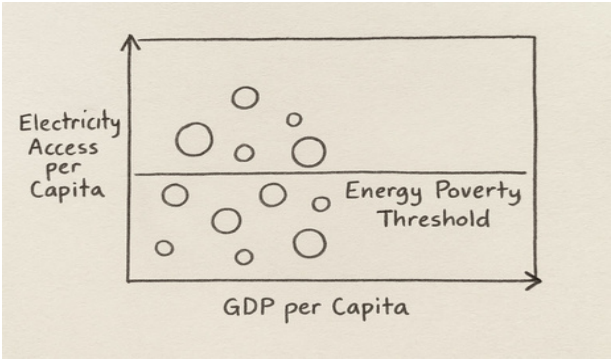
3. Visualization

3.1 Design Process and Initial Planning

The design process started by identifyingtheanalytical goal of this project: to compare several socioeconomic and energy-related indicatorssimultaneously in order to understand global disparities in electricity access. This led to the choiceofamultidimensional visualisation method, one able to represent several variables at once withoutburdening the viewer. Early exploratory sketches and rough drafts were helpful in evaluating optionssuch as scatter plots, bar charts, and line charts. These formats, however, could not meaningfullyaccommodate more than two to three variables.

To show this, a bubble chart is used becausethechart encodes multiple dimensions: position (GDP vs electricity), bubble size (population), andcolour(renewable energy share). This made it best suited for displaying inequality, cluster patterns, andoutliers in one go.

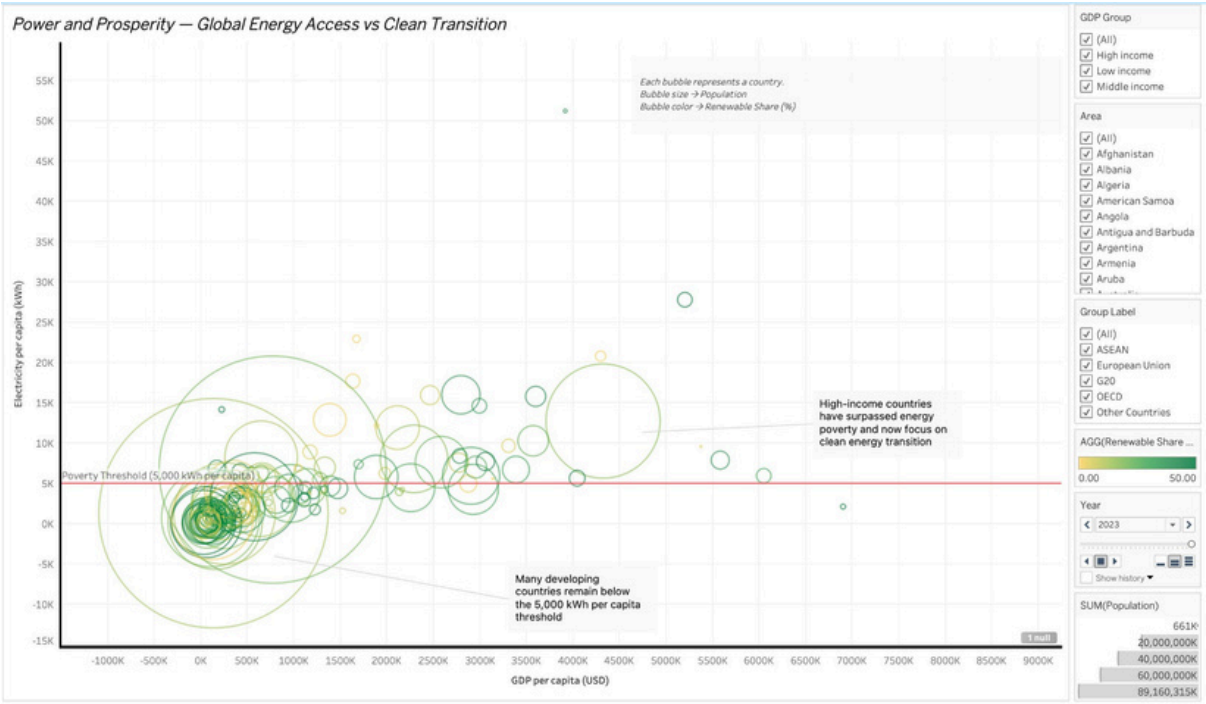
3.1.1 Sketch of Planned Visualisation



This sketch was used to guide the final layout: showing the intended arrangement of axes, energy poverty threshold line, colour gradient and the relative sizes of the bubbles.

3.2 Final Visualisation Output

The final bubble chart was then created in Tableau using the integrated dataset, with GDP per capita on the X-axis and electricity consumption per capita on the Y-axis. Population was encoded as bubble size and renewable energy share controlled the bubble color gradient. The 5,000 kWh Energy Poverty Threshold was added as a horizontal reference line to classify countries above and below adequate electricity access.



3.3 Choice of Chart Type

A **bubble chart** was chosen for several reasons:

- It displays four variables simultaneously, which is crucial for analyzing multidimensional problems such as energy inequality.
- It highlights the relationship between GDP and access to electricity, the core research question.
- The size of the bubble represents each country's size, allowing readers to compare economic and demographic capacities with electricity generation and understand how wealth and population affect energy production and distribution.
- Colour encoding shows patterns in renewable energy adoption, also providing further insight into transitions towards sustainability.

The chart type is therefore justified because it supports multivariate comparison, structural inequality mapping, and pattern detection across countries and income groups

3.4 Design Choices and Justification

- **Colour:** Sequentialgradient(light to dark) for renewable energy share, supporting perceptual ordering.
- **Shapes and Marks:** roundbubbles preserve perceptual equivalence; variation in area shows variationin population.
- **Layout:** On the X-axis theGDP is plotted, while on the Y-axis electricity per capita is plotted. Itseparates high- and low-access countries. A threshold line supports the interpretation of this figure.
- **Font and Labels:** Clean and minimal labels for readability.
- **Overplotting:** reduced opacity and filters G20, EU, ASEAN, income groups clean up clusters.

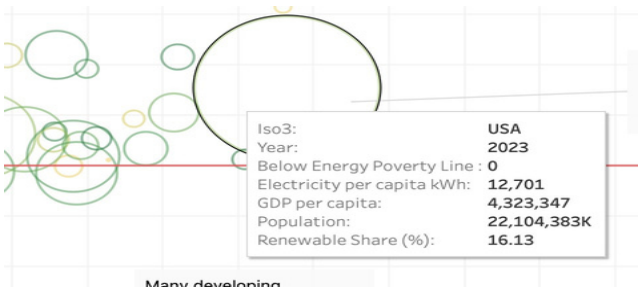
3.5 Interactivity and Animation

Enhance exploration and communication:

- **Tooltips:** Iso3, Year, Below Energy Poverty Line, Electrity per capita,GDP per capita, Population and Renewable Share.
- **Filters:** GDP Group, Area and Group Label.
- **Highlighting:** Clicking a bubble highlights a country for comparison.
- **Year:** Users can scroll through the selected years to visualize changes over time.

Each of these features allows for detailed analysis, micro-pattern discovery, and clear communication of global energy inequalities.

Filters



Tooltips

4. Conclusion

The project effectively examined world energy inequality by combining datasets about electricity generation, GDP per capita, population, and renewable adoption. The multi-variable bubble chart visualization presented how the state of national prosperity, demographic scale, and renewable energy progress together shape the state of energy access worldwide. The bubble chart was able to

graphically reveal patterns like the disproportionately high levels of energy poverty in countries with low income that are densely populated and the strong connection between economic development and reliance on renewable energy.

Tools and Libraries Used:

The above visualization was created with Python data analysis and plotting packages. More specifically, pandas was used to clean and merge data and The final bubble chart created in Tableau. Testing and iterative development was done in the Tableau environment; various other tools and libraries were examined but not integrated to preserve clarity and simplicity in the visualization.

Critical Reflection:

This final visualization aligns very strongly with the core research question through the clear depiction of global energy inequality in a multivariable bubble chart. The intuitive use of bubble size and color presents disparities across countries, while an energy-poverty threshold line provides a meaningful reference point for comparison. These design choices helped in keeping the visual story accessible and easy to interpret. Beyond the basic structure, the visualisation was strengthened by the interactive features implemented in Tableau. Specifically, users are able to view tooltips of the key metrics, apply filtering options to focus on groups of countries of interest, and scroll through years to observe temporal change. These affordances enable deeper engagement with the data than would be possible in a static plot and allow the audience to explore patterns in the data that might otherwise go unnoticed. However, there are some points where the visualisation still has room for improvements: a fully animated transition between several years, more sophisticated geospatial representations, or other customized interactions such as dynamic comparison views may provide even richer insights. Given the time and scope of the project, it was decided not to implement these enhancements, but they remain useful directions for future work.

Collaboration:

The project was done in collaboration by Vaseekaran Krishnan Vinodhan and Avinash Rajasekar. We have divided the work, based on our strength, supported each other throughout the data analysis and its visualization process, and discussed the progress and findings periodically. Effective communication and problem-solving jointly helped to make the workflow smooth and productive, hence assuring high-quality outcomes of the project.

References

1. Ember Energy. *Yearly Electricity Data*. Accessed November 2025. https://ember-energy.org/data/yearly-electricity-data/

2. World Bank. *Total Population (SP.POP.TOTL)*. Accessed November 2025. https://data.worldbank.org/indicator/SP.POP.TOTL?utm

3. Our World in Data. *Share of Population Living in Urban Areas*. Accessed November 2025. https://ourworldindata.org/grapher/share-of-population-urban?time=1960..2024

4. Our World in Data. *GDP per Capita (World Bank, constant USD)*. Accessed November 2025. https://ourworldindata.org/grapher/gdp-per-capita-worldbank-constant-usd?time=1960..2024

Tools and Libraries

Tableau Software. *Tableau Desktop*. Version 2025. https://www.tableau.com/ – Used for data integration, visualization, and interactive bubble chart creation.

Tutorials

1. How to track data evolution over time with Tableau Desktop - https://www.youtube.com/watch?v=IK6AE3u6LGs
2. Bubble Chart in Tableau - https://www.youtube.com/watch?v=RPDFuRjdfqg

Declaration on the Use of AI Tools: We confirm that AI tools (GPT) were only used for gathering **information about the project scope, refining visualisation concepts, and general data search queries**; all core **data analysis, graph design, and conclusions** were performed manually by our team.